

Uniform Convergence of a Kernel Estimator

(sample paper for the mobile review workflow)

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Abstract

We study a kernel-type estimator of a regression function and establish sufficient conditions for its uniform consistency. The analysis is efficient in the sense that it requires only finite second moments of the noise.

1 Introduction

Nonparametric regression is a workhorse of modern statistics. The difficulty of establishing uniform, rather than pointwise, rates is well known; see [1] for an overview. Our goal is a self-contained proof that avoids the usual entropy calculations.

Throughout, $X_1, \dots, X_n$ are i.i.d. with density f on $[0, 1]$ and $Y_i = m(X_i) + \varepsilon_i$, where the errors ε_i are independent of the covariates and satisfy $\mathbb{E}\varepsilon_i = 0$ and $\mathbb{E}\varepsilon_i^2 = \sigma^2 < \infty$.

2 Methodology

Let K be a bounded, compactly supported kernel and $h = h_n \rightarrow 0$ a bandwidth sequence. The Nadaraya–Watson estimator is

$$\hat{m}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x-X_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-X_i}{h}\right)}. \quad (1)$$

Theorem 1 (Uniform consistency). *Suppose m is Lipschitz, f is bounded away from zero on $[0, 1]$, and $nh/\log n \rightarrow \infty$. Then*

$$\sup_{x \in [0, 1]} |\hat{m}_h(x) - m(x)| \xrightarrow{p} 0.$$

Proof. The bias term is handled by the Lipschitz condition. For the stochastic term, a standard chaining argument over a grid of mesh h^2 combined with Bernstein's inequality yields the rate. We therefore conclude that the estimator is consistent uniformly over the unit interval, which is the sufficient condition required for the affine correction step in Section 3. $\square$

Lemma 1. *Under the assumptions of Theorem 1, the variance of the stochastic term is of order $(nh)^{-1}$ uniformly in x .*

3 Conclusion

We have shown that the kernel estimator in (1) is uniformly consistent under weak moment conditions. Future work will investigate whether the workflow extends to dependent data and to the affine efficiency correction of [2].

References

- [1] Evarist Giné and Richard Nickl. *Mathematical Foundations of Infinite-Dimensional Statistical Models*. Cambridge University Press, 2016.
- [2] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.